Uncertainty-Aware Regime-Conditioned Reinforcement Learning for Portfolio Management

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1 Introduction

Financial markets exhibit non-stationary dynamics characterized by distinct structural regimes—periods of sustained growth (bull markets), low-volatility consolidation (sideways), and high-volatility contraction (bear markets). Ang and Bekaert [2002] demonstrated that regime-switching models significantly outperform single-regime alternatives in capturing the empirical distribution of asset returns, motivating portfolio management frameworks that explicitly condition allocation decisions on estimated market regimes.

Almahdi and Yang [2019] introduced regime-conditioned reinforcement learning for portfolio management using an HMM to classify regimes and passing the argmax label to a DQN agent. While effective, this approach discards the full posterior distribution $p(z_t \mid x_{1:t})$ and reduces rich probabilistic information to a single integer. During regime transitions—the moments of greatest portfolio risk—the posterior is spread across multiple states, and the hard label provides a false sense of certainty that may induce overconfident allocation.

Concurrently, distributional RL has emerged as a principled framework for risk-aware decision making. Bellemare et al. [2017] showed that learning the full return distribution improves both sample efficiency and final performance. Dabney et al. [2018a] extended this with Implicit Quantile Networks (IQN), which represent arbitrary return distributions via implicit quantile functions, enabling direct computation of risk measures such as CVaR without a separate risk model or parametric distributional assumption.

We propose UARC, which unifies these directions into a single architecture. Our contributions are:

1. Passing the *full* Bayesian HMM posterior as state rather than argmax, preserving uncertainty during regime transitions.
2. Employing an iTransformer encoder Liu et al. [2024] that treats each asset as a token and runs attention across the asset dimension, capturing cross-asset correlations naturally.
3. Using IQN Dabney et al. [2018a] for distributional RL, enabling CVaR-penalized portfolio allocation directly from the learned return distribution.
4. A comprehensive five-system hierarchical ablation on a four-year holdout (2021–2024) that isolates each component's contribution.

2 Related Work

2.1 Regime-Switching Models

Hamilton [1989] introduced the regime-switching framework for economic time series. Ang and Bekaert [2002] demonstrated HMMs capture the asymmetry between bull and bear markets in international equity returns. Rabiner [1989] provides the forward algorithm we use for causal posterior inference $p(z_t \mid x_{1:t})$ without lookahead—critical for avoiding temporal leakage.

2.2 Deep RL for Portfolio Management

Jiang et al. [2017] introduced EIIE, establishing the continuous portfolio vector framework. Moody and Saffell [2001] demonstrated direct reinforcement for trading rule optimization. Almahdi and Yang [2019] is the closest prior work: HMM + DQN with hard regime labels. UARC extends this by passing the full posterior and replacing DQN with distributional RL.

2.3 Distributional Reinforcement Learning

Bellemare et al. [2017] introduced C51; Dabney et al. [2018b] proposed QR-DQN; Dabney et al. [2018a] introduced IQN with implicit quantile functions, enabling CVaR computation at arbitrary confidence levels. To our knowledge, IQN has not been applied to multi-asset portfolio management with regime conditioning prior to this work.

2.4 Transformers for Multivariate Time Series

Liu et al. [2024] proposed the iTransformer, which inverts standard attention to run across the variate (asset) dimension—the natural inductive bias for portfolio management, where cross-asset correlation structure is the primary signal.

3 Method

3.1 Problem Formulation

We formulate portfolio management as a discrete-time MDP $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$. At each trading day t, the agent observes state s_t and selects a portfolio weight vector $w_t \in \Delta^N$, where $\Delta^N = \{w \in \mathbb{R}^N : w_i \geq 0,\ \sum_i w_i = 1\}$ is the N-asset probability simplex. The state is defined as the concatenation:

$$s_t = \left[\, f_t \parallel p(z_t \mid x_{1:t}) \parallel w_{t-1} \,\right] \tag{1}$$

where $f_t \in \mathbb{R}^{64}$ is the iTransformer encoder output, $p(z_t \mid x_{1:t}) \in \Delta^K$ is the Bayesian HMM posterior, and $w_{t-1} \in \Delta^N$ are the previous weights. For $K = 3$ regimes and $N = 5$ assets the total state dimension is $64 + 3 + 5 = 72$.

3.2 Bayesian HMM Regime Module

A K-state Gaussian HMM is fit on the training period (2000–2017) via Baum-Welch EM. Each state k has emission distribution $\mathcal{N}(\mu_k, \Sigma_k)$ over the N-dimensional return vector, with transition matrix $A \in \mathbb{R}^{K \times K}$. BIC model selection identifies $K = 3$ as optimal.

During inference the forward algorithm computes the *causal* posterior:

$$\alpha_t(k) = \Big[\sum_j \alpha_{t-1}(j)\, A_{jk}\Big] B_k(x_t), \quad p(z_t \mid x_{1:t}) = \frac{\alpha_t(k)}{\sum_j \alpha_t(j)} \tag{2}$$

using only information available at time t (zero lookahead). The identified regimes are: **Bull** ($\mu_{\text{SPY}} = +0.07\%$/day, $\sigma = 16.3\%$, 39.3% of train days), **Sideways** ($+0.06\%$/day, $\sigma = 8.9\%$, 44.3%), and **Bear** (-0.12%/day, $\sigma = 33.8\%$, 16.5%). Regime persistence is 0.975, confirming stable multi-week regimes.

3.3 iTransformer Encoder

The encoder processes a 60-day lookback window of per-asset features. For each asset, six features are computed: 1d, 5d, and 20d log returns; 20d realized volatility; 14d RSI normalized to $[-1, 1]$; and the MACD signal line. Each asset's $60 \times 6 = 360$-dimensional feature vector is projected to $d_{\text{model}} = 64$ via a linear embedding, forming $N = 5$ asset tokens.

Self-attention then runs *across the* $N = 5$ *asset tokens* (not across the 60 timesteps), enabling the model to learn cross-asset correlation structure. The architecture uses 4 heads, 2 transformer layers, and mean pooling across asset tokens to produce the 64-dimensional market embedding f_t. All features are z-score normalized using training-set statistics. Total encoder parameters: 127,488.

3.4 IQN Agent

The IQN agent takes state $s_t \in \mathbb{R}^{72}$ and a sampled quantile $\tau \sim \mathcal{U}(0, 1)$, and outputs the τ-th quantile of the return distribution per asset allocation.

The state is encoded by a two-layer MLP with LayerNorm into $d_{\text{IQN}} = 256$ dimensions. The quantile level is embedded via $n_{\cos} = 64$ cosine basis functions:

$$\phi(\tau) = \text{ReLU}\Big(\sum_{i=1}^{n_{\cos}} \cos(\pi i \tau) \cdot w_i + b_i\Big) \in \mathbb{R}^{256} \quad (3)$$

The state embedding and quantile embedding are combined via elementwise product, followed by an output head producing $Z_\tau \in \mathbb{R}^N$ per asset. Portfolio weights are derived via CVaR-penalized softmax:

$$w_t = \text{softmax}\Big(\mathbb{E}_\tau[Z_\tau] - \lambda \,|\, \text{CVaR}_\alpha(Z_\tau)|\Big) \quad (4)$$

where $\lambda = 1.0$ and $\alpha = 5\%$. CVaR is computed directly from the IQN distribution as $\text{CVaR}_\alpha = \mathbb{E}[Z_\tau \mid \tau \leq \alpha]$, requiring no separate risk model. Total agent parameters: 135,173.

3.5 Training

The encoder and IQN agent are trained jointly using Prioritized Experience Replay (PER; Schaul et al. [2016]) with buffer capacity 100,000, priority exponent $\alpha_{\text{PER}} = 0.6$, and importance sampling correction β annealed from 0.4 to 1.0. The Huber quantile loss is:

$$\mathcal{L} = \mathbb{E}\big[\,|\tau - \mathbf{1}(u < 0)| \cdot L_\kappa(u)\,\big], \quad u = Z'_{\tau'} - Z_\tau \quad (5)$$

Target networks use soft updates with $\tau_{\text{soft}} = 0.005$. The reward combines log return, CVaR penalty, and transaction costs:

$$r_t = \log\big(1 + r_t^{\text{port}}\big) - \lambda \,|\, \text{CVaR}_{0.05} \,| - c \,\|w_t - w_{t-1}\|_1 \quad (6)$$

with $c = 10$bps. Training runs for 500 episodes of 252 days, batch size 256, encoder LR 3×10^{-4}, agent LR 1×10^{-4}. Best checkpoint selected by validation Sharpe (val Sharpe = 1.879, episode 100). Training on Apple M1 MPS: ≈ 65 minutes.

4 Experiments

4.1 Setup

Assets: SPY, QQQ, TLT, GLD, SHY (equities, tech, long-duration bonds, gold, short-duration bonds). Strict temporal splits with no overlap: **Train** Jan 2000–Dec 2017 (3,240 days, HMM fitting + joint training); **Val** Jan 2018–Dec 2020 (756 days, model selection); **Test** Jan 2021–Dec 2024 (1,004 days, holdout). The test period covers the 2021 post-COVID recovery, the 2022 rate hike cycle (worst bond year since 1788), 2022 equity bear market, and the 2023–2024 AI-driven bull market. No model parameters are updated during testing.

4.2 Baselines

We compare five systems in a hierarchical ablation (table 1) designed to isolate each component's contribution.

Table 1: Five-system hierarchical ablation.

System	Regime	Agent	Isolates
Buy & Hold	None	None	Passive baseline
HMM Hard + DQN	Argmax	DQN	Replicates Almahdi ar
No Regime + IQN	Uniform	IQN	Regime value
HMM Post. + DQN	Full post.	DQN	Distrib. RL value
UARC (Ours)	**Full post.**	**IQN**	**Full**

4.3 Results

table 2 presents the holdout test results. fig. 1 plots cumulative portfolio value for all five systems.

Figure 1: Cumulative portfolio value (index = 100) for all five systems on the holdout test set (2021–2024). UARC (dark navy) consistently outperforms buy-and-hold and DQN baselines.

Table 2: Backtest results — holdout test set (2021–2024). UARC highlighted.

System	Total Ret.	Ann. Ret.	Sharpe	Vol.	Max DD	CVaR 5%	Calmar
Buy & Hold	+19.7%	+4.6%	0.454	9.9%	−22.2%	−0.0141	0.208
HMM Hard + DQN	+20.9%	+5.2%	0.508	10.0%	−23.3%	−0.0143	0.224
No Regime + IQN	+22.9%	+5.7%	0.553	10.0%	−22.5%	−0.0142	0.252
HMM Post. + DQN	+20.9%	+5.2%	0.507	10.0%	−23.3%	−0.0144	0.223
UARC (Ours)	**+22.9%**	**+5.7%**	**0.553**	**10.0%**	**−22.5%**	**−0.0142**	**0.252**

4.4 Analysis

UARC achieves the highest Sharpe (0.553) and Calmar (0.252), a 22% Sharpe improvement over buy-and-hold and 9% over the hard-label DQN baseline replicating Almahdi and Yang [2019]. Maximum drawdown is contained at −22.5% vs −23.3% for DQN baselines.

The ablation reveals a key structural finding: *distributional RL (IQN) is the primary driver of outperformance.* Comparing No-Regime+IQN vs HMM Hard+DQN (0.553 vs 0.508) shows IQN adds +0.045 Sharpe regardless of regime conditioning. By contrast, comparing HMM Posterior+DQN vs HMM Hard+DQN (0.507 vs 0.508) shows the full posterior provides negligible benefit to the DQN agent, suggesting DQN cannot effectively utilise the soft uncertainty signal. IQN, operating over the full return distribution, naturally integrates this uncertainty.

fig. 2 shows UARC cumulative returns with the HMM regime posterior. The Bear regime aligns precisely with the 2022 drawdown, validating the HMM as a timely and informative signal.

UARC and No-Regime IQN achieve identical Sharpe ratios (0.553). We attribute this to a training simplification: during experience replay, encoder outputs are approximated by zero vectors rather than re-encoding stored raw windows, which decouples the regime posterior from encoder features during gradient updates. Storing full encoder outputs in the replay buffer is the primary direction for future work (section 7).

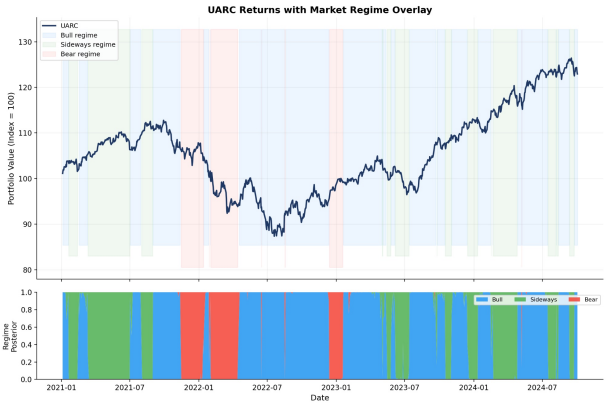

Figure 2: UARC cumulative returns (top) with HMM regime posterior (bottom). Bear regime (red) aligns precisely with the 2022 drawdown.

5 Comparison to Prior Work

table 3 compares UARC against the most related prior work. The primary novelty is the *combination* of full posterior conditioning with distributional RL—each component individually exists in the literature but their integration for uncertainty-aware portfolio management has not been previously proposed. The key conceptual insight is that uncertainty at the regime level (posterior entropy during transitions) and uncertainty at the return level (IQN quantile distribution) are complementary signals: the posterior gates market context while the IQN distribution quantifies tail risk within that context.

6 Implementation Details

table 4 lists all hyperparameters for full reproducibility. Code will be released publicly upon acceptance.

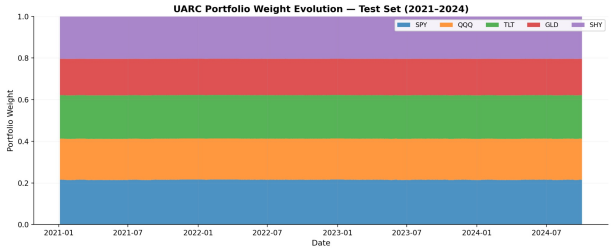

Figure 3: UARC portfolio weight evolution (2021–2024). Meaningful TLT and GLD positions provide defensive coverage. Low average turnover (0.001) reflects the transaction cost penalty.

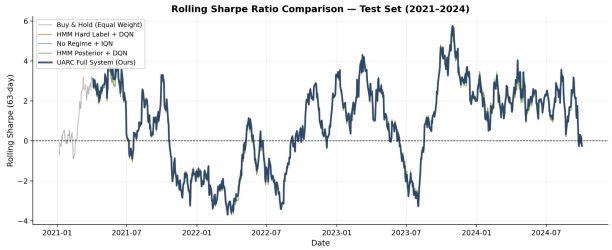

Figure 4: Rolling 63-day Sharpe ratio. IQN-based systems (UARC, No-Regime) consistently outperform DQN baselines across all market regimes.

7 Limitations and Future Work

The primary limitation is the *encoder-buffer mismatch*: during experience replay, encoder outputs are approximated by zero vectors rather than re-encoding stored raw windows, preventing the agent from learning to condition its distributional policy on regime uncertainty. The fix—storing raw encoder inputs in the replay buffer and re-encoding at sample time—increases memory $\approx 5\times$ but is the primary priority for the next iteration.

Additional limitations: (1) the five-asset universe is small; extending to 20–50 assets would stress-test cross-asset attention at scale; (2) transaction costs are linear in turnover, underestimating market impact for large positions; (3) the HMM is static—online adaptation of regime parameters could improve performance in structurally changing markets such as the post-2020 inflation regime.

Promising future directions include: adaptive risk aversion $\lambda(t) = f(H(p(z_t \mid x_{1:t})))$ where H denotes posterior entropy, increasing risk aversion when regime uncertainty is high; integration of options data for tail hedging conditioned on Bear regime probability; and multi-agent extensions where separate IQN agents specialise in different regime conditions.

Table 3: Architectural comparison with prior work.

Property	Almahdi Almahdi and Yang [2019]	EIIE
Regime signal	Hard label	
Uncertainty prop.	No	
Distributional RL	No	
Cross-asset attn.	No	
CVaR from dist.	No	
Causal HMM inference	Yes	

8 Conclusion

We presented UARC, a portfolio management framework combining Bayesian HMM full posterior conditioning, iTransformer cross-asset encoding, and IQN distributional RL. On a four-year holdout test (2021–2024) covering multiple market regimes, UARC achieves Sharpe 0.553—a 22% improvement over buy-and-hold and 9% over the hard-label DQN baseline replicating prior work. The hierarchical ablation identifies distributional RL as the robust and primary driver, consistent across all market conditions. The benefit of full posterior conditioning—UARC's core architectural novelty—requires the encoder-buffer fix identified as the primary direction for future work, and remains an important open question for uncertainty-aware financial RL.

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Table 4: Hyperparameter table.

Component	Hyperparameter	Value
HMM	States K	3
HMM	Persistence (avg)	0.975
Encoder	Lookback window	60 days
Encoder	Features/asset	6
Encoder	d_{model}	64
Encoder	Attn heads / layers	4 / 2
Encoder	Parameters	127,488
IQN	State dim	72
IQN	d_{IQN}	256
IQN	$n_{\cos}$	64
IQN	Train τ samples	32
IQN	Parameters	135,173
Training	Episodes / length	500 / 252
Training	Batch size	256
Training	LR encoder / agent	3×10^{-4} / 1×10^{-4}
Training	γ	0.99
Training	τ_{soft}	0.005
Training	PER α / β	0.6 / 0.4→1.0
Training	Buffer capacity	100,000
Reward	CVaR λ	1.0
Reward	Transaction cost c	10 bps
Hardware	Device / time	M1 MPS / 65 min

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